

Tanuj Dontaraju

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EDUCATION

University of Delaware

GPA: 3.8/4.0

B.S. in Artificial Intelligence Engineering

Expected Graduation: May 2028

Relevant Coursework: Calculus I–III, Data Structures, Discrete Mathematics I, Data Science I, Engineering Mathematics I, Introduction to Systems Programming, Introduction to Digital Systems

EXPERIENCE

University of Delaware - DIR Lab

Aug 2026 – Present

Research Assistant

- Examining early identification of I/O problems in HPC parallel file systems based on actual telemetry data collection in fault-injection test facilities.
- Scoping I/O-phase datasets to compare ML algorithms with small LLM classifiers.
- Studying how to detect the earliest point of failure in the I/O phase to balance decision speed and accuracy.

Mercor

Aug 2025 – Jan 2026

AI / Model Training Contributor

- Labeled and structured image and text datasets to directly support machine learning model training and fine-tuning.
- Developed data preprocessing pipelines that validated input data and removed inconsistencies before training began.
- Set high data quality standards across data pipelines to ensure reliable AI model performance.

PROJECTS

Verifi (Currently in Development)

React, TypeScript, TanStack Router, Supabase, Mobile Development

- Developing an application on the web and mobile that connects clients to bonded, background-checked independent and mobile notary publics.
- Creating a backend via Supabase to securely synchronize data from the website to mobile platforms.
- Programming responsive light/dark mode interfaces, a secure PDF upload system, and a location-based notary search for easier notarization access.

StudyLens (Personalized RAG Chatbot)

Python, Streamlit, LangChain, ChromaDB, LLM APIs



- Built a Retrieval-Augmented Generation (RAG) chatbot using Python and LangChain to answer questions directly from personalized course materials.
- Created a search pipeline using ChromaDB with fast PDF and text document ingestion.
- Developed a Streamlit interface utilizing OpenAI, Gemini, and Anthropic LLMs automatically.

F1 Dashboard

Python, React, Next.js, FastAPI, TypeScript



- Built a Formula 1 (F1) race telemetry dashboard (React/Next.js + FastAPI) with analysis tabs (Strategy, Pit Stops, Lap Analysis, Qualifying/Practice) and real-time race alerts.
- Implemented a glassmorphism UI with Apple design principles using spring animations and animated navigation highlights.
- Currently adding a news page with weekly F1 updates during race weekends.

SafeCircle (Mobile Safety Application)

TypeScript, REST APIs, Geolocation, AI Integration



- Collaborated on a three-person team during the University of Delaware's Hen Hacks hackathon to develop a mobile safety application with live location tracking and emergency communication.
- Integrated generative AI and speech-to-text APIs to automatically record and summarize audio during live emergencies.
- Built a backend REST API to support location tracking and audio processing.

SKILLS

Programming Languages: Python, JavaScript, Java, TypeScript, HTML, CSS

Technologies: React, TanStack Router, LangChain, LLMs, PyTorch, TensorFlow, Retrieval-Augmented Generation (RAG), Git, REST APIs